COLLEGE OF INFORMATION AND COMPUTING SCIENCES

UNIVERSITY PHILOSOPHY
The University serves the INDIVIDUAL by providing the student with a nurturing environment for optimal human flourishing. It serves the community by offering programs responsive to individual and societal needs.

VISION
CSU is a University with global stature in the arts, culture, agriculture and fisheries, the sciences as well as technological and professional fields.

MISSION
Cagayan State University shall produce globally competent graduates through excellent instruction, innovative and creative research, responsive public service and productive industry and community engagement.



SYLLABUS IN IT ELECTIVE 2

Course Credits	**3.0**	**Semester**	**First**	**School Year**	**2024-2025**	**Number of Lecture Hours**	**36**
Pre-requisite/s	**NONE**		**Co-requisite/s**	**NONE**		**Number of Laboratory Hours**	**54**

Course Description:	This course focuses on the study of the fundamental concepts in programming languages which includes major tools and techniques. This includes syntax specification, binding and scoping; arrays and list; control structures, procedural abstraction and parameter passing. The course will demonstrate on exploring key characteristics of major programming paradigms, including their relationship to the imperative programming paradigm adopted by mainstream programming languages.
Intended Graduate Attributes	**The Cagayan State University aims to produce graduates who will exhibit:** **COMPETENCE:** • Critical thinker • Creative problem solver • Competitive performer • Regionally • Nationally • Globally **SOCIAL RESPONSIBILITY:** • Sensitive to ethical standards • Steward of the environment for future generations • Social justice, and economic equity advocate.

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	UNIFYING PRESENCE: ● Uniting theory and practice ● Uniting strata of society ● Unifying the nation ● The ASEAN region, and the world ● Uniting the University and the community.
Course Intended Learning Outcomes (CILO's)	**At the end of course, students are expected to:** **CILO 1:** Design develop and test a program that uses a messaging service that sends asynchronous messages across the network **(IT03,IT04,IT05,ITO6)** **CILO 2:** Design, develop and test a program the uses SAX or DOM to parse an XML document, XSL and XSLT to transform a data stream from one format to another. **(IT03,IT04,IT05)** **CILO 3:** Write, debug and test a script using an operating scripting language to facilitate the management of an operating system.**(IT03, IT04,IT07,IT08)** **CILO on Internationalization:** Demonstrate the ability to integrate diverse programming languages, frameworks, and platforms cohesively within a single application**.(IT13)** **CILO on Gender and Development, and other SGDs:** Apply method for collecting and analyzing data that take into account gender-specific needs and preferences.**(IT11,IT12)**
Grading System	As per memorandum order VPAA 5105-MEMO-2022-09-47 from the Office of the Vice President for Academic Affairs dated September 12, 2022, regarding the blended learning grading system, the following shall be the bases of the student's grade: A. Academic Subjects Submission of assigned student activity(online & on-site) 20% Student engagement/forum discussion(online & on-site) 25% Quizzes and Unit Examination 25% Major Examination (Midterm and Finals) 30% B. For Subjects with Lecture and Laboratory Lecture grade (follow the above grading system) 60%

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	Laboratory grade	40%
	Total	100%

Time Frame	Learning Content	Intended Learning Outcomes (ILOs)	Teaching and Learning Activities (TLAs)	Learning Assessments	Teaching and Learning References/ Resources	Modality
Week1	University Vision, Mission, Goals and Objectives, College ILO, Program ILO, College Policies, Grading system Course Requirements	At the end of the session, students are expected to: Demonstrate knowledge of University's VMGO, and College policies and guidelines, targets, grading system and course requirements of the subject	**Lecturette** Jingle Composition	**Quiz** Presentation of the composed jingle	Resources / References Administration Manual, Academic, Manual, Student, Manual CMO of the Program/s	Hybrid

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Week 2-3	**Introduction** a. Overview of Programming Languages b. Aspects of programming language features c. Importance of studying programming languages d. Concept on compilers and run-time systems e. Comparison between and among programming languages f. Installing programming language.	At the end of the session, students are expected to: • Define programming languages • Enumerate the different features of programming languages • Compare and contrast the different programming languages • Know the Advantages and disadvantages of using programming languages • Understand how to Install and set up the working environment	Discussion Presentation of Officials Audio-Visual presentation	Quizzes Graded Recitation	www.w3schools..com www.geeksforgeeks.org	Hybrid
Week 4-6	Common syntaxes, Variables and Data Types a. Common syntaxes b. Variables c. Data types d. Date and time String	At the end of the session, students are expected to: • Enumerate and explain the different commons syntaxes • Understand the roles and uses of variables • Explain the different Data types	Presentation Discussion Lecture Hands-on activities	Quizzes Graded Recitation	www.w3schools..com www.geeksforgeeks.org Sestoft, P, PROGRAMMING LANGUAGE CONCEPTS, Second edition,2017	Hybrid

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		• Understand the use of each Data type. Format date, time and strings				
Week 7-9	**Control Structures** a. Conditional Statements b. The If elif else statement The Switch statement	• Understand the use of the different conditional statements • Be able to used the structures in solving basic to advanced problems.	Presentation Discussion Lecture Hands-on activities Demonstration	Quizzes Activity	www.w3schools..com www.geeksforgeeks.org Sestoft, P, PROGRAMMING LANGUAGE CONCEPTS, Second edition,2017	Hybrid
Week 10-12	**Iterations** a. Loop structure b. Looping procedures Looping techniques	• Understand the different ways on how to construct loops • Create ascending and descending loop Implement multi layered looping structure.	Presentation Discussion Lecture Hands-on activities Demonstration	Activity Graded Recitation	www.w3schools..com www.geeksforgeeks.org Sestoft, P, PROGRAMMING LANGUAGE CONCEPTS, Second edition,2017	Hybrid
Week 13-14	Arrays a. One-Dimensional Arrays b. Sorting and Searching in an Array c. Two-and-Multidimensional Arrays	• Create and understand the structure of a single dimension array • Implement some processes used in manipulating arrays like sorting, searching and indexing. • Create and understand the	Presentation Discussion Lecture Hands-on activities Demonstration	Quizzes Graded Recitation Activity	www.w3schools..com www.geeksforgeeks.org Shaw, Zed A, LEARN PYTHON 3 THE HARD WAY, 3rd Edition, 2016	Hybrid

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		structure of a multi-dimensional arrays. • Create programs using the techniques in managing arrays Create programs using the techniques in managing arrays				
Week 15-16	**Classes/Function** a. basic class b. Class with parameters c. Class with return values d. Complex Class	• Create class using the simplest method • Create classes/functions with one or more parameters. • Create classes/functions with return values. Create classes/functions using subclasses as components.	Presentation Discussion Lecture Hands-on activities Demonstration	Quizzes Graded Recitation Activity	www.w3schools..com www.geeksforgeeks.org Shaw, Zed A, LEARN PYTHON 3 THE HARD WAY, 3rd Edition, 2016	Hybrid
Week 17-18	**File Management** a. Managing External file b. Reading and writing to a file c. Appending and Deleting content of a file.	• Access external file like text file to the program. • understand the different methods in reading and writing to an external file. • Recognize the different commands in appending and deleting of an external file.	Presentation Discussion Lecture Hands-on activities Demonstration	Quizzes Graded Recitation Activity	www.w3schools..com www.geeksforgeeks.org Shaw, Zed A, LEARN PYTHON 3 THE HARD WAY, 3rd Edition, 2016	Hybrid

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INTENDED LABORATORY ACTIVITIES (if applicable)

1. Develop a program that employs the various operation including arithmetic, logical, relational and conditional operations.(week 4-6)
2. Design a program that utilize Python's flow control mechanisms.(week 7-9),(week 10-12)
3. Develop a program that utilizes various operators with Python arrays'(week 13-14)
4. Design a python program that demonstrates the concept of classes and objects.(week 15-16)

OTHER MAJOR COMPETENCIES OUTPUT (i.e., Reflection Paper/Portfolio/Film Review, etc. *if applicable*)
Aside from the major examinations, the student will be assessed at other times during the term by the following:

- NONE

RUBRIC FOR ASSESSMENT TO BE USED FOR Major Competencies Outputs (MCOs) (if applicable)

Criteria	Excellent 20 pts.	Good 15 pts.	Satisfactory 10 pts.	Unsatisfactory 5 pts.	Score
Functionality	The program functions flawlessly, meeting or exceeding all requirements	The program generally functions as expected, with minor issues or one major issue	The program partially functions, but significant functionality issues are present.	The program does not meet basic functionality requirements.	
Code Quality	Code is exceptionally clear, well-organized, and follows best practices.	Code is mostly clear and well-organized, with a few areas that could be improved.	Code organization and clarity could be significantly improved.	Code is messy and difficult to follow.	
Efficiency	Code is highly efficient and optimized for performance where necessary.	Code is generally efficient, with a few areas where optimizations could be made.	Code could be made more efficient in several areas.	Code is inefficient and may have significant performance issues.	

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Documentation	Thorough and well-written comments and documentation are provided throughout the code.	Adequate comments and documentation are included, but some areas lack detail.	Minimal comments and documentation are provided, making the code difficult to understand.	No comments or documentation are included.	
Creativity	Innovative solutions and approaches are evident throughout the code.	Some creative solutions are implemented, enhancing the overall quality of the code.	Few creative elements are present in the code.	The code lacks creativity and relies heavily on standard solutions	
Testing	Innovative solutions and approaches are evident throughout the code.	Test coverage is sufficient but may lack thoroughness in some areas.	Testing is limited and does not adequately cover all functionalities.	No testing or inadequate testing is performed.	
Total	16-20	11-15	6-10	5	

REFERENCES

Print

- NONE

Non-Print

- www.python.org/(https://docs.python.org/3/)

Software

- Visual Studio Code
- IDLE
- SPYDER

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COURSE POLICIES
(on classroom norms, i.e., attendance, missed exams, academic integrity and medium of communication (provided that they don't run counter to existing policies)

- Students are expected to attend classes regularly. If the student accumulates absences for at least 20% of the total number of class hours, the student shall be considered dropped. Dropping should be done before the midterm examination to be considered officially dropped, otherwise, considered failed (except in cases found meritorious).
- The medium of communication shall be English, except in those courses where the use of the national language is necessary.
- Students must adhere to the mandates of academic integrity. Academic dishonesty (cheating, plagiarism, etc.) is taken very seriously. The minimum penalty for a violation of academic integrity is a failure (zero) for the assignment. When using course materials or other credible sources during class discussions and written requirements these must be cited properly.
- Use of electronics during class is not allowed. Turn off/mute/set to vibrate any electronic devices before class begins. For personal emergency, the student asks to be excused from the class to communicate outside of the classroom.
- Missing an examination is not allowed. In extreme cases such as death in the family, sickness or accident, a written evidence is required to make up a missed exam. No make-up quizzes shall be allowed.
- Students are encouraged to contact their professor for consultation and feedbacking at the following contact information:

[Name of Professor]
[Place of Work]
Consultation Hours:
Contact No. (Optional)

Prepared by:

BSIT FACULTY MEMBERS

Reviewed by:

Program Coordinator/Department Chair or College Dean

Endorsed by:

GENERINO P. SIDDAYAO, DIT
University Dean

Recommending Approval:

LIRIO GUDINA C. MANGAWIL, Ph.D
Director of Instruction

Approved by:

MARIDEN V.CAUILAN,DPA
Vice President for Academic Affairs